

Bill of Material

<i>Component Name</i>	<i>Quantity</i>	<i>Cost Per Unit (ZAR)</i>
1:10 Scale RWD On-Road Chassis	1	R 850.00
ESP32-WROOM-32U Microcontroller	1	R 145.00
2.4GHz External IPEX Antenna	1	R 25.00
TF-Luna Micro LiDAR Distance Sensor	1	R 450.00
Sharp GP2Y0A21YK0F IR Sensor	2	R 180.00
3-Pin JST-PH Wire Harness	2	R 12.50
DS3218MG 20kg Digital Servo	1	R 290.00
25T Aluminum Servo Arm Horn	1	R 35.00
3650 Brushless Motor + 60A ESC Combo	1	R 450.00
2S 7.4V 2200mAh 30C LiPo Battery	1	R 220.00
LM2596 DC-DC Buck Converter	1	R 45.00
Heavy Duty Inline Toggle Switch	1	R 25.00
Latching Red E-Stop Button Switch	1	R 45.00
PETG Filament (Sensor Mounts)	1	R 40.00
M3 Stainless Fasteners & Nuts	1	R 35.00
18 AWG Silicone Power Wire Pair	1	R 20.00
Breadboard Jumper Ribbons (M-F/F-F)	1	R 25.00

FINAL GRAND TOTAL COST

Is (Component Breakdown & Direct Links)

<i>Total Cost (ZAR)</i>	<i>Supplier Name</i>	<i>Supplier Stock Code</i>
R 850.00	HobbyMania	HM-RWD10-CHS
R 145.00	Micro Robotics	MR-ESP32-32U
R 25.00	Micro Robotics	MR-ANT-2G4-IPX
R 450.00	Communica	COM-TFLUNA-LID
R 360.00	BotShop	BS-SHARP-2Y0A21
R 25.00	BotShop	BS-JST-PH3P
R 290.00	DIYElectronics	DIY-SRV-DS3218
R 35.00	DIYElectronics	DIY-ACC-25THRN
R 450.00	HobbyMania	HM-3650-60A-CMB
R 220.00	HobbyMania	HM-LIPO-2S2200
R 45.00	Micro Robotics	MR-PWR-LM2596
R 25.00	Communica	COM-SW-TOG-12V
R 45.00	Communica	COM-SW-ESTOP-RED
R 40.00	In-House Print	3DP-PETG-BLK
R 35.00	Fastener World	FW-M3-KIT-SS
R 20.00	BotShop	BS-WIR-18AWG-PR
R 25.00	BotShop	BS-WIR-JMP-SET
R 2,990.00	COMPLIANT	UNDER BUDGET

Direct Purchase URL / Link

hobbymania.co.za/chassis-110-rwd
microrobotics.co.za/esp32-wroom-32u
microrobotics.co.za/ipex-antenna-2g4
communica.co.za/tf-luna-lidar
botshop.co.za/sharp-ir-sensor
botshop.co.za/jst-ph-3pin-cable
diyelectronics.co.za/ds3218mg-servo
diyelectronics.co.za/25t-servo-horn
hobbymania.co.za/3650-60a-combo
hobbymania.co.za/lipo-2s-2200mah
microrobotics.co.za/lm2596-buck
communica.co.za/toggle-switch-12v
communica.co.za/e-stop-button
diyelectronics.co.za/petg-black
fastenerworld.co.za/m3-kit-ss
botshop.co.za/18awg-silicone-wire
botshop.co.za/jumper-wire-set

[Budget Cap: R 3,000.00]